Year 12 Maths - Using graphs

Similar questions - Regions defined by inequalities

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The curve C has equation $y = x^2 - 2x - 3$ and the line L has equation $y = x - 3$.

(a) Find the coordinates of the points where C and L intersect. **[4]**

(b) Sketch C and L on the same axes, marking the intercepts of C with the axes. **[3]**

(c) On your sketch, shade the region R for which $y \geqslant x^2 - 2x - 3$ and $y \leqslant x - 3$. **[1]**

(d) State, with a reason, whether the point $(1, -3.5)$ lies in R. **[2]**

2 The region R is the triangle with vertices $O(0,0)$, $B(6,0)$ and $A(2,4)$ (including its boundaries).

(a) Show that A lies on both the line $y = 2x$ and the line $x + y = 6$. **[2]**

(b) Write down three inequalities that define R. **[3]**

(c) Find the area of R. **[2]**

(d) How many points with integer coordinates lie in R (including its boundaries)? **[3]**

Answers

Question 1

(a) $x^2 - 2x - 3 = x - 3$, so $x^2 - 3x = 0$ and $x = 0$ or 3. Points $(0,-3)$ and $(3,0)$

(b) C is a U-shaped parabola crossing the x-axis at $(-1,0)$ and $(3,0)$, the y-axis at $(0,-3)$, with minimum $(1,-4)$. L is a straight line through $(0,-3)$ and $(3,0)$

(c) R is the finite region between the parabola (below) and the line (above), from $x = 0$ to $x = 3$

(d) At $x = 1$: $x^2 - 2x - 3 = -4$ and $x - 3 = -2$. Since $-4 \leqslant -3.5 \leqslant -2$, the point is in R

Question 2

(a) $y = 2x$ gives $4 = 4$ and $x + y = 6$ gives $2 + 4 = 6$

(b) $y \geqslant 0$, $y \leqslant 2x$, $x + y \leqslant 6$

(c) $\frac{1}{2} \times 6 \times 4 = 12$

(d) Counting by x: $x = 0$: $(0,0)$ (1 point); $x = 1$: $y = 0,1,2$ (3); $x = 2$: $y = 0,\ldots,4$ (5); $x = 3$: $y = 0,\ldots,3$ (4); $x = 4$: $y = 0,1,2$ (3); $x = 5$: $y = 0,1$ (2); $x = 6$: $y = 0$ (1). Total 19